

RENANCE

AMBROSE ALLI UNIVERSITY, EKPOMA

CSC 303 — Past Questions & Materials

by Renance

Course: CSC 303

Institution: Ambrose Alli University, Ekpoma

Papers in this pack: 1

Levels: 300

This study pack was compiled and restructured by Renance for CSC 303 students at Ambrose Alli University, Ekpoma. It brings together 1 past paper(s)/material(s) into one clean, readable document so you can revise without chasing scattered PDFs. Practise these same questions interactively on Renance with instant answers, AI explanations, and full CBT mock exams — over 235,000+ questions across JAMB, Post-UTME, WAEC, NECO and university courses.

Study online: renance.app

Get the app: Google Play — [dev.renance.renance](https://play.google.com/store/apps/details?id=dev.renance.renance)

Generated by Renance AI

Paper 1 of 1 — CSC 303

Level 300 • First Semester • PastQuestion

VACULTY OF PHYSICAL SCIENCES

AMBROSE vd UNIVERSITY, EKPOMA een q

DEPARTMENT OF COMPUTER SCIENCE ' |

hs hc i First Semester Degree Exe nination 2020/2021 Academic Session /

mputer Systems Architecture Fime: 2'/44hre Date: 22/12/2021 }

\sfRucr ' ,

r a! !ON: Answer Question | and any other three. j

jon 1

BV hat is} omputer Systems Architecture? 2 Mar

The mes Common goals in computer architect fe rev olve around the tradeo!

perform@ce. Do you agree? Just ly your claim) 3 marks

What is th primary motive for introducing Boolean Algebra? 2marks _ |

Simplify th following expressions using Boolean A igebra (i) $AB + ABC + ABC$ (ii) $AB Y A(CD + CD$

> (ii) $(BE AND \backslash AB + CD)$ 2 marks each A ,

) Given the jaar! function = $xyz \#XyzP xy$ (i) Write the truth table of (he fun@uion (11) Draw the

n algebra (iii) Simplify the algebraic expression and draw the lox

is betwçten cost and

ogic diagt 'p using the original Boolea

Pm Siavrim Yorirks each ep

Rye xVF1 XyZ 1X7 1 xyg (i) Write the truth table of the function (ii

plily the algebraic expression usit

'Juestion 2

) Given the Boolean function |

Draw the logie diagram using the original Boolea aigebga (iii) Sim

.-map and draw the logic diagram. 2 mark each 4

Jotermine the minimal form of an expression given by $U A, B, C, 1D) = \& (1, 2-3, 5, 7, 8, 9, 10, 1$

©) Using a suitable illustration, proof that "if an algebraic unction has the sum of n products, thé physical gate

niust have n number of AND gates and one OR gate. Gimarks i

A

: os

hn the Boolean expression for |; and simplify using a

Question 3 | erie

Represent the following functions on a truth table,

wap) $KAS A, B, C, D) > IL(0, 1, 2, 4, 8, 9, 1)$ 3 marks each

Rin and state their properties? 4 marks

1G] GD) = (1,3,4,5,7,10,13, 14) ei

ndard product of suas. 3 HarkS

15) 4marks

plement the AND and OR functions with a logical

<) Rewrite the expipssion $f(x,y,z) = WR + AV$ es in

) Write short note} on the following (i) Words (ii) Nibble (iii) Byte 1 mark each

4 Micstion 4 }

\\ hat are the basic information processing functions of the digital computer s3 stem? 2 marks

)What are binary Jogies? What are the three distinctive properties of a logical variable? 4 marks be

) Represent the follow ing function on a K-Map and obtain she simplified Boolean expression

$F(A,B,C)$ 2013.10

jaw the NAND-NAND gate structure for the simplified, function. 6 marks

se diagrams to describe the Von Neumann Archilecture and the System Bus Model.4 marks

Question 5 .

4) Convert 1001111119.110100101 to (i) Octal (ii) Hexadecimal 2 marks each

(b) Describe the two major categories of the switching or logical function? 3 mark

fc) Draw the NANIDD-NAWD gate structure for the function $F(A,B,C,D) = AD + BC + BCD$ 3 marks

(d) Draw up a truth table showing the output from a 4-input logical industrial process cont-oller operating as

follows: output is high when at least one of the four inputs A, B, C, and D is high, but, output is low whol i

either A and C or A and D are high together, 4 marks ~#

Question 6 f

(a) Represent the funetion $f(A,B,C,D) = \sum(1,3,4,5,6,7,9,12,15)$ on a K-Map, obtain the simplified Boolean

expression for and Draw the NAND-NANI) gate structure for the simplified expression, 6 wuarks f

of © 'b) Given the truth table below. Write out the simplifcd Boolean algebraic term 4 marks

XY!

00}0 Oe

01 }1 6

og N Dy (al

IT yi -.

ic) Convert the hexadecimal number 18715 to () binary and (i) octal, 2 A A

} 4 Cac

t

~ (d) Hluswate the levels of machines 2 marky i O| {

hye. t QEIES t Nae 2}

loa NC ree

da ji

dint oe

to al Col Le pu
 4 i 09 aN Cy ey
 \ iif Neo —, e (eSS a a: an
 tA thO sate \,d re
 ' (c C\ AG
 aes
 Var a es
 td oS Be Vac = ae Jo Me bya aly be!
 sh ay } B, av eal bach ava sole S Hd
 bash iy a | and © “They 4 Leet |,
 ow
 eperebes Eni, Pe No}
 eo i rc Vor able i
 a ioe R039 of
 (0) E (hR6) = £, bosutieD
 ye!
 CIN6| ecard Ws Leer ©
 OSE ALLI UNIVERSITY, EKKPOMA
 TYSICAL SC WENCES
 COMPUTER SCIENCE
 2019/2020 Session
 Date. 08/03/2021
 AMBR
 i}ACULTY OF PI
 DEPARTMENT OF
 First Semester Degree Examination
 TIME: 2'; hrs
 jomputer Architecture
 CSC 303: ©
 on
 INSTRUCTION: Answer Questi
 Question 13
 (a) Comptiter sy!
 (b) Use diagrams
 land any other three
 tecture is not about using computers to design building. Discuss? — 2 marks
 Neumann Architecture and the System Bus Model. 4 marks
 jucing Boolean Algebra'? 2 marks__

Boolean Algebra (i) $AB + ABC + ABC$ (ii) $AB + A(CD + CD)$

stems Arch

to describe the Von

(c) What was the primary motive for intro

Simplify the following expressions using

(iii) $(BC + AD)(AB + C)$ 2marks each :

(d) Given the Boolean function $f = xy + xz + yz$. (i) Write the truth table of the func

logic diagram using the original Boolean algebra (111) Simplify the algebraic expression and dra
a 2 marks each

Function (ii) Draw the

w the logic

evolve around the tradcoffs between cost and

diagram. ;

J (e) The most common goals in computer architecture

F performance. Do you agree? Justify your claim. 2marks \

Question 2 &

anization? 2 marks

n for F and simplify using

(b) Represent the following functions on a truth table, obtain the Boolean expressio

ak-map; (i) $F(A,B,C,D) = \sum (1,2,3,4,7,10,12,13)$ (ii) $F(A,B,C,D) = \sum (0, 2, 8,10,$

(c) Memories are all random access. Discuss? 3 marks

(d) Using asuitable illustration. proof that 'if an algebraic function has the, sum of n products, the
ph

must have n number of AND gates and one OR gate. 3marks :

(a) What ts Computer Org

12,14) 4 marks each

ysical gate structure

Question 3 «

(a) What does the term "Memory Access

(b) Convert 100111] 1100.110100101010 to (i) Octal (

+ the AND and OR functions with alogical system and state their properties? 4 murks

a K-Map and obtain the simplified Boolean expression $F(A,B,C)$ -

for the simplified function. 6 marks

Time" mean? 2 marks

ii) Hexadecimal 2 marks each

¥(1.3.4.5.7).

(o) inp tn

(d) Represent the following function on

Draw the NAND-NAND gate structure

Question 4

- (a) What are the basic information processing
- (b) Determine the minimal form of an expression g_i
- (c) Given the truth table below. Write out the simplified Boolean functions of the digital computer system? 2 marks
- ven by $F(A,B,C,D) = \sum (1, 2, 3, 5, 7, 8, 9, 10, 12, 15)$ 6 marks
- lean algebraic term. 4 marks

XY iE

00/1 . o|®D

o1}i of} }

10 {0 ahs

11 {0 \

- (d) Convert the hexadecimal number F879%5 to (i) binary and (ii) octal. 2 marks each
- ; th

Question 5

- (a) What is the design objective of memory hierarchy? 2 marks
- (b) Write short notes on the following (i i
- ying (i) Instruction set (ii i i
- : set (ii) Addressin > (iii icG
- . Deen ghd gate structure for the function $F(A,B,C,D) + 05_{10} = \text{baa pr bel ce}$
- eben tres i seien ie Aap from a 4-input logical industrial process controller o is He
- ; our inputs By ase is hi . ' A ee A
- A and D are high together. 4 marks puts A, B, C, and 1) is high, but, output is low ue aiehA take
- and C oi

Question 6

- (a) What are binary logics? What are isti
- é s] 2 the three distinctive pr ics of ¢ i
- bs short sai on the following (i) Words (ii) Byte gp pe ee
- ermine the minimal form of the expression g_i ' GUL
- sn : |! pression given by $F(A,B,C,D) = \sum 3, 5, 7$
- (d) Describe the two major categories of the switching or logical oe fon peers
- ? 4 marks

a -

mone ae 7 ee pig ; gee

: = xt {KvOMA! Ne ot NS

PINT YRRSEUY ENCES ' . _

MBRODE ALL ae TURAL SCL

AN oe cacemnttkoess

' _ _ —

reve vires' lo (OS Ba ee

: KPOMA

MBBROSE ALLI UNIVERSITY, E

a FACULTY OF NATURAL SCIENCES

DEPARTMENT OF COMPUTER SCIENCE

First Semester Degree Examination 2013/2014 Session

TIME: 2', hrs

CSC 303; Computer Architecture :

INSTRUCTION: Answer question one(1) and any other three

i \

tion 1 .

Vi How would you describe Computer Architecture?' 3 marks

(b) Write informed note on the following (i) K-map (ii) duality relation (iii) logic gates aioe eae me
ACY) NOTE Cl aw : \ ; : 'cro ace

(c) Use a simple diagram to describe a microprogram representation in ri roe aa Se physcal

¥¢) Using a suitable illustration, proof that 'if an algebraic fiinction has the-sum i ,

2 OR gate. 4 marks a

te structure must have n number of AND gates and one OR g¢ 4 . +c

ae) Simplify the following expressions using Boolean Algebra (i) $AB + ABC + ABC$ (ii) $AB + A(CD)$
- @i)(BC+ADXAB+CD) —. 3 marks each

vestion 2 ;

'~a) What are memory caches? Why are they required in the computer industry? 4 marks

sib) Rewrite the expression $F(A,B,C) = A \oplus B \oplus C$ in standard sum of products. 4 marks

(c) State the following theorems (i) Duality theorem (ii) De Morgan'stheosem 4 marks each

Question 3 ' . ;

(a) What are the characteristics of RISC-machines? 3 marks

(© Implement the AND and OR functions with a logical system. Write their algebras and properties?
6 marks

(c) Draw the NAND-NAND gate structure for the function $F(A,B,C) = D(4,2,5,7)$.5 marks

(¢) Represeat the following function on a K-Map and obtain the Simplified Boolean expression
 $F(A,.BO\} = \sim\sim\sim$

L245, 5) dmgrhs. _ Ea eS oe

oe et ee fee : in ae ea ae

"Question 4 :

a) What is the design objective of memory hierarchy? 4 marks

(©) Simplify the function given by $F=AY+AZ+YZW$ 5 marks

(c) Explain any' two possible ways of exuréssi

any two possible ways of expressing logical functions. In other words, use illustrations to show how one can move from one form to another. 6 marks - 2 marks to show why

(c) How does the operating system relate to the memory? 3 marks

{Question 5 as well

(a) What are binary logics? What are the distinctions, i.e.

three distinctions?

(b) What makes an exclusive OR different from the parallel addition?

(c) Using suitable example, illustrate the logical relationship between variables. 5 marks

Define the minterm, the maxterm and the truth

(d) Draw the NAND-NAND gate structure for

the function $F(A, B,$

$C, D) = AD + BC + BCD$

Question 6. } ee

in a pie chart, draw the logic diagram, 4 marks

~~) Means of a table, the valid =

, $ABC = A + E + C$ demonstrates De Morgan's theorem for three variables:

3 What is an instruction? 4 marks . .

(c) Convert the hexadecimal F

decimal F87D5 to binary and octal. marks

P

> AAT

5s Uie

CSC 303 — Compu

INSTRUCTION:

Question i

(a) Computer Architec

(d) Describe the two

; : 1

(e) Rewrite the express

i

(f) Determine the minin

AD illustrate the levels <

Question 2°) |

a) What are the ct

wee we S

(b) A majority gate with
case of 4 inputs
input is b. Simplify

c) What does the term

d) Convert 100181157

'Question 4:

a) What are the necessary

(b) Represent the following

NAND gate symbol

c) What is the relationship:

3) Write short notes for

i

Question 5 :

(a) Describe any. Y or S

(b) What is the purpose of

'- Draw the NAND.

/ Draw up a truth table

output is high! why?

A and B are high

Question 6 :

(4) Write short notes on

) Given the Boolean

logic diagram, using

logic simplify,

c) Convert the hex

4) What are the basic

2. 1,2,3,4,7,10,12,13) (i) $F(A,B,C,D) = \sum (0, 2, 8, 12, 14)$ 4 marks each

SD gate structure for the function

AMBROSE ALLEN UNIVERSITY, IKPOMA

SCHOOL OF PHYSICAL SCIENCES

DEPARTMENT OF COMPUTER SCIENCE

Final Semester Degree Examination

2016/2017 Academic Session

Systems Architecture

Time: 2 Hours Date: 05/05/2017

Answer Question I and any other three. So \

e is not about using computers to design building. Discuss? 2 marks each

Modern computer architecture revolves around the tradeoffs between cost and performance. Do

5) im. 3 paarks

i Von Neumann Architecture and the System Bus Model. 4 marks

gior categories of the switching or logical function? 4 mark

: GS FOW,X,Y,2Z) = WX +XY + YZ in standard product of sums ' '

it) F(W,X,Y,Z) = XOY@ Z in standard sunt of products 2 marks each

form of the funcison F(W.X,Y,Z) = WXYZ + XYZ + WY + W '

"machines 3 marks

be tf

y in the design of the jogic circuit? 3 marks i

ure provides an output of logic level one when a majority of the inputs are at log:c | ;

v,2Z, write a logical expression for a variable F which is equal to | whene /OF, a majo

expression and draw thé gate structure using AND and OR gates 6 rarky

Zemory Access Time" méan? 3 murky '

- 119100101010 to Gi) Ogtal (ii) Hexadecimal 3 marky each

(Arid)

able. obtain the Boolean expres

rasization? 2 marks

functions ona Goth

ion for Vand simp "l¥ using a S-map;

'form of an expression given by 'F(A,B,C,D)= 2 (1,2,3,5,7,10,12,) 4x:

» suid to have temporal and spatia! locality? 4 marks —AGD

s

i

i '

*y sicps to construct a full adder? \$ marky

g function on a K-Map and obtain the simplifid Boolean expression. Draw the NAND-

simplified function. (A ,B.C,D) = DC 3.4.5.7, 10, 12 M15). Sp carks

FyiVy \$2, 4

tween the clock frequency and the clock period? = 2 marks

. € following (i) Words (ii) Nibble Gifi) Bytel 2 marks each

uenlia! Circuit types indicating cach of their circuit diagrams? 6 maré&s

wie system clock? 2 marks

ion. P(A,B,C,R) = AD + BG + BCD 4 marks :

lowing the output from a 4-input logical industrial process controller o

least one of the four inputs A,B. Cy and D is high, bute

ace. Using a k-map, weite out the sir

pers ting as follows:

alpat is low when e'ther A aad Cor
uplifcd Boolean expression, 6 nrarky

2 following (i) tnstruction s

z \ t (il) Addrressing mode (iii) Logie G

ti $XYL + XYZ + XYZ$

Rye XyZz (i) Write the trun t

ii) Simplily the aly

Me 2 marcy each

able of the funeti

vbraic expression using a k

"inal Boolean algebra (

8H) Draw the

marks each

Nap cad draw the

2 Marky cach

a! computer system? 2

4 dumber E87D5 to (i) Binary and (ii) Octal,

4 processing Lunctions of the digit

; 2 et aS

Nf Ro] oA du te

OES:

Vith relevant examples briefly explain the three fundinentel

>fulness ia digital sys'em development (6, marks)

Design the logic cire it for the following:

g wns

j = SZ D DD) +2 marks) ee | .

) $[A*(4+B)D].C$ (2 marks)

iii) $XY(CHDYBLCK X+Y)$ (2 marks): °

sa. What is switching algebra? Explain how

| ae cellphone Hes ign 7 omer

output based on the curr 'ent inputs and the:

fs mest of its prope:ties' and advice |

marks)

c. Describe « register. States the primary t

parallel i ane s and outputs (6 marks) .

of operati

ms fered if the digital item must

">, Explain what you understand by
 Briefly describe how the system is designed
 Design the circuit logic using (i) AND OR
 of the sum of products (6 marks)
 based)

Binary algebra and State its usefulness
 Design a system that will produce an
 output: Construct this system with detail
 possible challenges and their solutions (7
 marks)
 Design a register and design a register with
 processor model. with its properties and
 the signals, that must be
 the PRN and the address bus,

AMBROSE ALLY UNIVERSITY, EKPOMA
 FACULTY OF PHYSICAL SCIENCES
 DEPARTMENT OF COMPUTER SCIENCE

First Semester Degree Examination 2018/2019 Academic Session

CSC 303 ~ Computer Systems Architecture ; Time: 2 hrs Date: 30/05/2019

INSTRUCTION: Answer Question 1 and any other three.

Question 1

(2) The most common goals in computer architecture revolve around the tradeoffs between cost and performance

Discuss, 7 marks

(b) Convert 0011111100.110100101010 to (i) Octal (ii) Hexadecimal 2 marks each

(c) Use diagrams to describe the Von Neumann Architecture and the System Bus Model. 4 marks

(d) Describe the two major categories of the switching or logical function? 3 marks

(e) Rewrite the expressions (i) $F(W,X,Y,Z) = WX + XY + YZ$ in standard product of sums

(ii) $F(W,X,Y,Z) = X \oplus Y \oplus Z$ in standard sum of products 2 marks each

(f) Determine the minimal form of the function $F(W,X,Y,Z) = WXYZ + XYZ + WY + W$ 4 marks

Question 2

(a) What is Computer Organization viz a viz computer architecture? 2 marks

(b) Represent the following functions on a truth table, obtain the Boolean expression for F and simplify using a k-map

(i) $F(A,E,C,D) = \sum (1,3,4,8,7,10,12,13, 14)$ (ii) $F(A,B,C,D) = \sum (0,1, 2,4, 8,9,10,12,14)$ -4 marks each

(c) Determine the minimal form of an expression given by $F(A,B,C,D) = \sum (1,2,3,5,7,8,9, 10, 12,15)$ 4marks

(d) Convert the hexadecimal

number E87DS to (i) Binary and (ii) Octal. 2 marks each

Question 3 ~

(a) What are ϕ

(b) Represen

= Necessary steps to construct a full adder? 5 marks and ;

2 "ollowing function on a K-Map and obtain the simplified Boolean expression. Draw the NAND-NAND g: e for the simplified function. $E(A,B,CD) = \sum(1,3,4,5,7,8,9,10,12,14,15)$. 5 marks

{c) What is the onship between the clock frequency and the clock period? 2 marks

(4) Write short soves on the following (i) Instruction set (ii) Addressing mode (iii) Logic Cae 2 marks each

Question 4 a oP

(2) Memories er- all random access. Discus? 4 marks . a ae . %,

(5) Given the Box can function $F = k\%z + .xyz + xyz t + kyZ + xyZ$ (i) Write the truth table of the function (ii) Draw th

logic-diagrarr using the original Boolean algebra (iii) Simplify the algebraic expression using a k-map and draw

logic diagram 2 marks each a a ec ee

(c) When is a con sutation said to have temporal and spatial locality? 3 marks :

(ϕ) Using a suitat ϕ illustration, proof that 'if an algebraic function has the sum of n produts, the physical gate stru

must haven r amber of AND gates and one OR gate. 3marks ;

Question §

(a) Use asimpie iegram to describe a micro

(b) Use diagra:

(c) What was

é program representation in a microprogrammec. system. @ marks

0 describe the Von Neumann Architecture and the System Bus Model. 4 marks' f

imary motive for introducing Boolean Algebra? 2 marks

(ϕ) Simplify the : o!!owing expressions using Boolean Algebra (i) $AB + ABC + ABC$ (ii) $AB + A(CD + CD)$

Gil) $(BC + ADYAB + C)$ 2marks each sl

Question 6 :

(a) Briefly discus the differences between the microprogrammed control and the hardwired control 4 marks

(b) What is the p: rpose of the system clock? 2 marks

c) Draw the NA D-NAND gate structure for the function $F(A,B,C,D) = AD + BC + BCD$ 4 marks

(dj Ae represent the anction $F(A,B,C,D) = \sum(1,3,4,5,6,7,9,10,12,15)$ ona K-Map, obtain the simplified Boolean

expression fo F and Draw the NAND-NAND gate structure for the simplified expressica, 6 marks

iu

i

AMBROSE ALLI UNIVERSITY, EKPOMA
FACULTY OF PHYSICAL SCIENCES
DEPARTMENT OF COMPUTER SCIENCE

' First Semester Degree Examination 2017/2018 Session °

Ic 303; Computer Architecture TIME: 2 hrs Date: 06/04/2018

f SCRUTION: Answer Question 1 and any other three .

ys

i bats Computer Sysiems Architecture? 2 marks

a

iagram: rh i Bus Model. 4 marks

di s to describe the Von Neumann Architecture and the System E A é, _

e) aietS the following expressions using Boolean Algebra (i) $AB + ABC + ABC$ (ii) $AB \# A(CD + CD)$
() ye.

(iii) $(BC + AD)(AB + C)$ 2marks each oe o oe ;

(@) Given the Boolean function $F = xyz \ xyz + kyz$. (i) Write the truth table of the function (ii)
Simplify the

_ algebraic expression and draw the logic diagram. .2 marks each

(e) Simplify $F(A,B,C,D) = ABCD + BCD + AC + C$ using a:K- map 4 marks

Juesti 2 ; . .

«) Beielly discuss the diiferences betweeii the microprogré mmed control and the hardwired control
4 marks

(b) Simplify the following functions using a k-miap; (i) $F(A,B,C,D) = \odot (1,2,3,4,7,10,12,13)$. .

Gi) $F(A,B,C,D) = IT (0, 2, 8,10,12,14)$ 3 marks each

(c) What are memory caches? Why are they required in the computer industry? 5. marks . 7

~ (a) Using a suitable illustration, proof that 'if an algebraic function has the sum of n products, the
physical

gate structure must have n number of AND gates and one OR gate. 3marks :

Question 3 F

(2) Disenss the memory aierarchy based on its design objectives 2 marks

(6) Convert 10011111100.110100161010 to (i) Octal (ii)-Hexadecimal-- 2 marks each

(¢) When is a coniputation said:to have temporal and spatial locality? 4 marks a8

(¢) Represent the following function on a K-Map and obtain the simplified Boolean expression
 $F(A,B,C) =$

2(1,3,4.5,6). Draw the NAND-NAND gate structure for:the simplified function. 6 marks '

Question 4 :

~ () Memories are all random access. Discus? 4 marks '

(o Write short notes on the following (i) Words (ii) Byte' 2 marks each :

(c) Determine the minimal form of an expression given by $F(A,B,C,D) = .2(1,2,3,5,7,$

) Determine the minimal f express 9,10,12,14) 4marks .

(c) Implement the AND and OR functions with a logical system and state their properties? 4 marks

Question 5 .

(2) Briefly explain the factors that contribute to the access time

(9) Write short notes on the following (i) Instruction set (ii) Ad

(c) Draw the NAND-NAND- gate structure for :

(d) With suitable example, illustrate 'the

4 marks ' ;

of a memory system? 2 marks

(dressing mode (iii) Logic Gate 2 marks each

for the function $F(A,B,C,D) = AD + BC + BCD$ 4 marks

logical relationship between the Minterm, Maxterm and the truth table

Question 6

(a) Why are most primary or main memories of contemporary computers constructed from semiconductor components irrespective of the weaknesses of the semiconductor Storage? 3 marks —. anaes

(b) Determine the minimal form of the function $F(wxyz) : f$

, = WIFE + whe + WE + w 5 mark

(c) Given the truth table below. Write out the simplified Boolean algebraic term, 4 marks °

01/9

10|90

ii|1

(d) Convert the hexadecimal number E87D5 to (i) binary and (ii) octal, 2 marks each

vA

ie

fps kl" | CRO

| AMBROSE ALLI UNIVERSITY, EKPOMA

| FACULTY OF PHYSICAL SCIENCES

DEPARTMENT OF COMPUTER SCIENCE

First Semester Degree Examination 2019/2020 Session

-sc 303: Computer Architecture TIME: 2!/ , hrs Date: 08/03/2021

INSTRUCTION: Answer Question 1 and any other three

Question 1 . — . 7

if Computer systems Architecture is not about using computers to design building. Discuss? — 2 marks

Wo) Use diagrams to describe the Von Neumann Architecture and the System Bus Model. 4 marks

(c) What was the primary motive for introducing Boolean Algebra? 2 marks_ ,

Simplify the following expressions using Boolean Algebra (i) $AB + ABC + ABC$

iii) $(BC + AD)AB + C$ 2marks each lel ges

@ sd the ie hee ee $F = xyz + xyz$. (i) Write the truth table of the function (ii) Draw the

logic diagram using the original Boolean algebra (iti) Simplify the algebraic expression and draw the logic

diagram. 2 marks each

(e) The most common goals in computer architecture rev performance. Do you agree? Justify your claim. 2 marks

(ii) $AB + A(CD + CD)$

olve around the tradeoffs between cost and

Question 2

(a) What is Computer Organization? 2 marks — — i

(b) Represent the following functions on a truth table, obtain the Boolean expression for F and simplify using

ak-map: (i) $F(A,B,C,D) = (1,2,3,4,7,10,12,13)$ (ii) $F(A,B,C,D) = 1 (0, 2, 8,10,12,14)$ 4 marks each

(c) Memories are all random access. Discuss? 3 marks

(d) Using a suitable illustration, proof that 'if an algebraic function has the sum of n products, the physical gate structure

must have n number of AND gates and one OR gate. 3marks

Yv Question 3

(a) What does the term "Memory Access Time" mean? 2 marks

(b) Convert 1001111100.110100101010 to (i) Octal (ii) Hexadecimal 2 marks each

(c) Implement the AND and OR functions with a logical system and state their properties? 4 marks

(d) Represent the foliowing function on a K-Map and obtain the simplified Boolean expression $F(A,B,C) = \Sigma(1,3,4,5,7)$.

Draw the NAND-NAND gate structure for the simplified function. 6 marks %

Question 4

(a) What are the basic information processing functions of the digital computer system? 2 marks

VV (b) Determine the minimal form of an expression given by $F(A,B,C,D) = \Sigma(1,2,3,5,7,8,9,10,12,15)$ 6 marks

(c) Giver the truth table below. Write out the simplified Boolean algebraic term. 4 marks

XY|F

00 {1

0141

10 0

11 J0

(d) Convert the hexadecimal' number F87D5 to (i) binary and (ii) octal. 2 marks each

Vv Question 5 :

(a) What is the design objective of memory hicrarchy? 2 marks

(b) Write short notes on the following (i) Instruction set (ii

(c) Draw the NAND-NAND gate structure for the function F(

(d) Draw upa truth table showing the output from a 4-in output is high when at least one of the four inputs A

A and D are high together. 4 marks

Question 6

(a) What are binary logics? What are the three distinctive properties?

; y , Properties of a logical variable'

(b) Write short notes on the following (i) Words (ii) Byte (iii) 2 vast non ORs

©) Determine the minimal form of the expression given by $F(A,B,C,D) = (1,2,3,5,7, 10,12)$ fsiard

'd) Describe the two major categories of the switching or logical function? 4 marks rr a 7

) Addressing mode (i) Logic Gate 2 marks each

$A,B,C,D) = AD + BC + BCD$ 4 marks

put logical industrial process controller operating as follows:

» B,C, and D is high, but, output is low when either A and Cor

2009/2010 Wiest Semester Degree Examination

Pw

AWW vade

KY

% GH Ob

<i ff 7% ' 7

AMBOSE ALLE UNIVERSPEEY, 1 KroMA. "

_ FACULEY OF NATURAL SCIENCES |

DEPARTMENT OF COMPUTER SCIENCE

Times 2 Ys hours

SAQA: Conny afer Architecdire:

seoemever GaesGon ft andany other' questions. —

8) Whetis computer Architecture? (4 marks) 4 :

Ay Dracus the vs Wious Liyers af Computer Architecture with respect fo machine |
ization CEO miarksy

(2) othe followings thearens:

CW uality (heoren (ii) Race ore theorem (iD ocliuelumney (G. marks)

(ht List ond brcty explain any 2 types of logic design (6 marks) .

watis) iy ad captiatn any lh fretors Ut may alte cl ihe wehitecture ofa fab Oe :

- Ong

ata diagram, describe the Von Neumann Architecture. - :

Shin A\rehiweture different from other somputing machines'? a

ie ianarks)

a(b) Phe cost Coammon poals in computer Architecture roves around the trade
of Coleen cast and perfor: mauree, Do you agree? Justify your asstion,

aetlae'N

Wag.

55 the differences between the microprogs ammed control

4 : davired control (QO marks)

thy Wh are the features of a Complex Instruction Set Computer «c iscy?

reason(s) Uaat tid to the deaign of Reduced fastruction f Sct COIS?

Giv

(a2 oaarks)

(a) Why

ea

iG ive

soerits (10° Fark .)

E raat suitable example, illustrate the logical relationship, between the

Mines, Mastery and the truth table (12 marks) ; F:

(Wr aw ups wuth (able showing the output front a 4 = input louie industrial

yoperating. as follows:

high, but ouiputis Low when either A and C ara and D are

ther (idinarks) : aa SE

0 (a) Briel oexplain the YoHowing CONCEP hs

G) Mipeli stag Oy fastragtiény UI (ii) Register (ivy Simply Instrue tion Sel C orn uter

(SICS) (Cv) Lope yates (13 marks)

(b) liniplovaient thie AND and OR function with a foyical systemi, Write their

aly inand propertits (7aiarks 8)

Good Luck! A

"Cty, . . , "\

7

exe)

ANE SE ALL] UNIVERSITY, ERPOMA

DEPARTRI ae

Virst Sernveste of rere OF COMPUTER SCIENCE

Computer Systems Architecture won 220/203 Asses

ie

PRUC TION: Answer Question Land any other three

what js Computer ae Architecture? 2 marks >

| Phe most common goals in computer architecture revelv ag

5 ee Do you agree? Yustiy vouesttoee a cond i

Bint ADKARATD) 2earks each ihe)

+g Given the Boolean function = xv" URyz } xyz. (i) Write the truth table of ae fan

jogie diagram using the original Boolean algebra (iii) Simplify the algebraic © res

diagram, — Jmarky each

Question »

Given the Boolean function $F(x,y,z) = x^2y + xy^2z + x^2yz + x^2y^2z + x^2y^2z^2 + x^2y^2z^2 + x^2y^2z^2 + x^2y^2z^2$ (i) Write the truth table

Draw the logic diagram using the original Boolean algebra (iii) Simplify the algebraic

K-map and draw the logic diagram, 2 marks each

(b) Determine the minimal form of an expression given by $F(A,B,C,D) = \sum (1,2,3,5,7,8,9,10,11,12)$

(c) Using a suitable illustration, prove that if an algebraic function has the sum of n products, the function must have n number of AND gates and one OR gate. 6 marks

Question 37

(a) Represent the following functions on a truth table, obtain the Boolean expression for P and

K-map: (2) $F(A,B,C,D) = \sum (1,3,4,5,7,10,13,14)$ (i) $F(A,B,C,D) = \sum (0,1,2,8,9,10)$

(b) Implement the AND and OR functions with a logical system and state their properties? 4 marks

(c) Rewrite the expression $I(w,x,y,z) = w^2t + x^2v$ in a standard product of sums. 4

(d) Write short notes on the following (i) Words (ii) Nibble (iii) Byte 1 mark each

digital computer system? 2 marks

a logical variable? 4 marks

Boolean expression $F(A$

Question 4

basic information processing functions of the

computer What are the basic

(b) What are binary logics? What are the three distinctive properties of
and obtain the simplified

(c) Represent the following function on a K-Map

Draw the NAND-NAND gate structure for the simplified function. 6 marks

Architecture and the System Bus Model.4

(b) Use diagrams to describe the Von Neumann

Question 5 <

(a) Convert 1001111110_2 to (i) Octal (ii) Hexadecimal 2 marks each

(b) Describe the two major categories of the switching or logical function? 3 marks

(c) Draw the NAND-NAND gate structure for the function $F(A,B,C,D) = A^2B + B^2C + C^2D + D^2A$

(d) Draw up a truth table showing the output from a 4-input logical function. The function follows: output is high when at least one of the four inputs A, B, C, and D is high, but neither A and C or A and D are high together. 4 marks

Question

(a) Represent the function

expression for F

(b) Given the truth table

(a) Convert the hexadecimal

function. Illustrate the levels of

006%4.

ninety i COSE ALLL UNIVERSITY, EKPOMA ,
DEPA eras PART-TIME DEGREE PROGRAMMES
JA) Ktest rnc a "NT OF COMPUTER SCIENCE
: ver Degree Examination 2010/2012 Session

(0303: Computer Acchitecture TIME

' IME: 2 hes Date: 9" August, 2012

ysTRUCTION: Ansy or Question Cand any other three . . :

gestion 1°

How would you desc ihe the arehitecture ofa computer system? 2 marks & V

"oy Write informed note on the following (0 logic gates (ii) duality relation (iii) K-map 2 marks

3) Whatowas the primary motive for intraducing, Boolean Algebra? 2 marks Co en ow =

(ly Use a K-map to simplify the expression $F(pyqy,2) = PU G24 Py +p$ 6 marks be

e) Using a suitable i ral

iwiratian, prool thal "Can algebraic function has the sum of i Me

tion, pebraic (G ag sum of n products, the physical p:

- must have number of AND yates and ong OR gate. 2 marks er

§ Given the Boolean (aiehod Ho Nyy txt fxg) Write the truth fable of the function (ii) Draw the loxie/
diagram using the original Boolean algebra (iii) Simplify the algebraic expression and draw the logic
7

diagram. 7 narks a :

iS Hh yyau

Question 2 - = *

a) Use a simple diagrams to describe a micraprogram representation in a microprogranimed
system, Md marks

" «hy Simplify the following function in gum of products form by means ofa K-map. Draw the logic
diagram

with G) AND-OR gates (ii) NAND yates only FCA,BYC) = 21,2, 4,5,7) 7 aarks >

; wants eT age

(c) State che following heorems (1) Duality (theorem (ii) De Morgan's theorem 2 aarks cach i
Questions. - See : ry . "+.

(n) Write a tnt table fora full adder and draw the logic diagram. Sanarks Ge

(j) Determine by means ofa truth table, the validity of DeMeryan's theorem lor three variables:

; ABC: BIC: Aamarks

(ce) Convert the hexadecimal number FSGS to (i) binary and (ii) octal. 3 marks zach

J Question 4. 5 ae 4 i

(a) What are binary fojies? What are the three distinctive properties of a logical variable? NN 5
marks _

(is) Using suitalic essayple, ilustrate the logical relationship between the minterm, the maxterm
and the truth table anaes

* Me) Draw dhe NAND-NANID gate stracture for the fincben ABCD) = AD FBC BCD fmarks

- Question §

" — isthe desiyor objective of memory hicrwehy?. JF marks .,

tc) \ clon giver by Fo AY UE AZ YZW 4 marks *. RY

(c) Paplain any Ovo possible ways of expressing, logical funeti i ' .

: a pressing, Jagicat funetions. In other wor SC strations io show how one en?

move lenrwinedaAivie andijurs eh htekS L rovords, use illustrations fo shew how ons ean

) Question 6 e iG s

(a) Implement thd ANO and OR fonctions with a logical sve i

. ! LAE Ms ogical s¥stetn, Write their alpebras and properties' "7 /

(bh) Draw the NANI NAMD gale Struchire forthe function TIA B,C) = (1,2 \$7) Piiaabaigg fe wv

tel bad Wek hod bel } . rf WW ote

od

(\ (6) Represent the Folly ving fibion on ah Map and obtain the simptitiod Reale expression F(A RC)

202,45, 6) 6aarky Vv

Ww ; i ot oe of

' — ©

AME,

ve t AY CTE] BI ot OL at Po

teeny a ' — _—

'Dep p 3

ART ,

es AMBIRO SK NT COMPUTER SCILNCE :

tk MOKATR PANE. UNIVERSITY, EKPOMA' '

WA Sines tet LIME DEGREE PROGRAMMES | ik

C308. Computer AV itevtine Ms Pi ynrination 2U08/2009 Session + "

PRUCTION; Attempt all Oestiong , TiMKe 2 he i : i

SOY wd re Ne ! ¢:

. . 4 a Se fon A ond any three each from S<etions B and 0,' . 7

"Gn) What ry COMPU arch Teetune) © 'ean J i2 marks , i \$ in

"(by Lapham an

Pe oe Y He pos sible vw

"a A

show how one can nave hom one ee

Le aT

"(c) Caven the Guth bible by law, Write

ARE nt ae Boolean alyebraiv term

'. au f Feat & i

'.? SA heh ve

= f T1070 he / ahids

3 ; nia a te | ; en

: (a) Hi We Bf i

ie heres LAWN Yu, , ; f j te Ep!

) Wye. j way Aa ak obtained froma truth fable as alternative combinations:

k rhs ' ,

Seat s bie a 'yf Prosaf that kw. yy ahen swyrt wry + Xyz ft wye st 'xy | J wae : Vi

: Bia nicite: Hie Ins sal eghatnhip fe lorme the mitten, | the mavrernd und the Goth

) Seclina ht Re. 2d, snarky

i res Y ;

at it thes the iutevnilly: jorge? aed itifgnie itary rater sliflercet ctr other ¢ toulating, 5 desl =e

fey geet? Pyrite tepsluality diced (4) Rave Hazard' Theorem GH)

Bee ater 2 lig Linawh tv rf est i i

h(i. WND.OR Hat \$ ati) wan Wt is A

has ine wil i

~ SAARC 1) 2 80,2809, 1 1

Tih FLL emeine why nea ofa uth able. the validity 'of De fora': 0

ag, qi)! Alc PALES

(i) Using, DeMorgan' 's Theorems, stow that Ae Ti ' "ALS i 52

wore for three variables, Z

2 — (e) Canvert the hexedceimal nambar PRA TE u 16 fi) bins uy cad (ii) ocli.

K eae > { :

? Section © © 24 marks ee i

te e oy ia! speci!) the fancton (1) ARC Dy AG CDE ABCD + ABCD » ABCD asaminicrin ,

as 'iy FA, BCD) - (ALB CUD) Aehie Ey Ae C+D YA BIC 1 DMAE ef C+) as a maxicrm !

Pops a (b) Kewnte she expressiats (i) HOW 7) Lie F xy)(vt2) in standard sual products | * =

(i) FLA ILC) = Ants Cita tC) in standard product of sums Vis

(tc} What we Injic gays Ten sielly cn the three feaste: iy(pes

{d) sw VHA. bd) AUCLY nie a TAC TE : ;

(f) fe): Fie: most conm@op foals ji computer eho rote around the tradeoffs between cost and!

¢ py aruiaieg. 1a Yor ape! 2D Justify your clay ef

ly

Tk nee

"au

Rage |

Pory